MS^{2}A^{2}Net: Multiscale Self-Attention Aggregation Network for Few-Shot Aerial Imagery Segmentation

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***Abstract*— Few-shot aerial imagery segmentation refers to the task of segmenting specific objects in scenes that have not been encountered during training with a small amount of annotated data for reference. However, most existing few-shot segmentation algorithms are primarily designed for natural images, and there is still a lack of exploration in the context of remote sensing aerial imagery. In this article, we propose a novel multiscale self-attention aggregation network (MS^{2}A^{2}Net), dubbed MS^{2}A^{2}Net, to address the challenge of few-shot aerial image segmentation in terms of scarce data and network architecture. Specifically, we first incorporate the designed asymmetric momentum contrastive learning (AMCL) into the pre-training stage, to improve the representation capability of the backbone without the expensive labeled data. Then the frozen encoder is transferred to the downstream few-shot segmentation task as the feature embedding. In terms of network architecture, we design self-attention aggregation in multiscale feature fusion, to construct the dual correlation of foreground and background between support and query features at the pixel level. Besides, the coordinate attention is designed to rearrange the distribution of feature importance in both horizontal and vertical spatial order perspectives, which facilitates adaptive fusion with the multiscale features. To verify the availability of the proposed MS^{2}A^{2}Net, we also reconstructed two novel datasets dedicated to few-shot aerial image segmentation, called DLRSD-4^i and iSAID-4^i. The experimental results show that our approach MS^{2}A^{2}Net is superior in three few-shot benchmark aerial imagery segmentation datasets, which achieves competitive segmentation performance. Extensive ablation experiments also reflect the effectiveness and scalability of the proposed components and overall network architecture.**

***Index Terms*— Aerial imagery segmentation, few-shot learning, self-attention mechanism, self-supervised learning (SSL).**

I. Introduction

SEMANTIC segmentation aims to identify regional blocks of different semantic categories to obtain a pixel-level semantically labeled segmented image [1]. With the progress of remote sensor technologies and the remarkable development of deep neural networks, semantic segmentation has been widely explored in remote sensing fields, such as change detection [2], building extraction [3], landslide detection [4], and road extraction [5].

Although the reinforcement of deep learning has made encouraging progress in semantic segmentation, it usually relies on a large number of densely annotated images for training [6], [7]. The collection of sufficient labeled data is time-consuming and laborious. Besides, it is difficult for the deep model to generalize in unseen target categories [7], [8], [9]. The expensive annotation costs and poor network generalization performance drive the requirement for better model generalization using only a small number of labeled samples.

In response to the aforementioned request, few-shot semantic segmentation focuses on imitating humans to segment a novel concept in the unseen categories with only a few samples as a reference [10], [11], [12]. Recently, numerous excellent few-shot semantic segmentation studies [13], [14], [15], [16], [17], [18], [19] have made encouraging progress on natural images. However, due to the considerable differences between the aerial and natural images, the existing few-shot segmentation work is still to be developed in the field of remote sensing. We have summarized the following main reasons in conjunction with some representative cases in Fig. 1.

1) *Incompleteness of Few-Shot Aerial Image Segmentation Datasets:* In the training of few-shot learning, a large number of labeled samples are required to improve the generalization ability of the model with only a small number of sample references. The natural image annotation categories and the number of samples are abundant, such as the widely used PASCAL-5^i [20] and COCO-20^i [21] datasets, which are sufficient to support the training requirements of few-shot segmentation. However, the development of few-shot aerial image segmentation is constrained by the absence of dedicated datasets due to the limited number of labeled categories and samples. Currently, this dilemma is gradually being improved with the release of large-scale aerial segmentation datasets [22], [23], [24] and the improvement of research dedicated to few-shot aerial image segmentation [25], [26]. Nevertheless, there are only a handful of published few-shot aerial image segmentation datasets.
2) *High Variability in the Perspective of Segmentation Targets:* Most of the existing excellent few-shot

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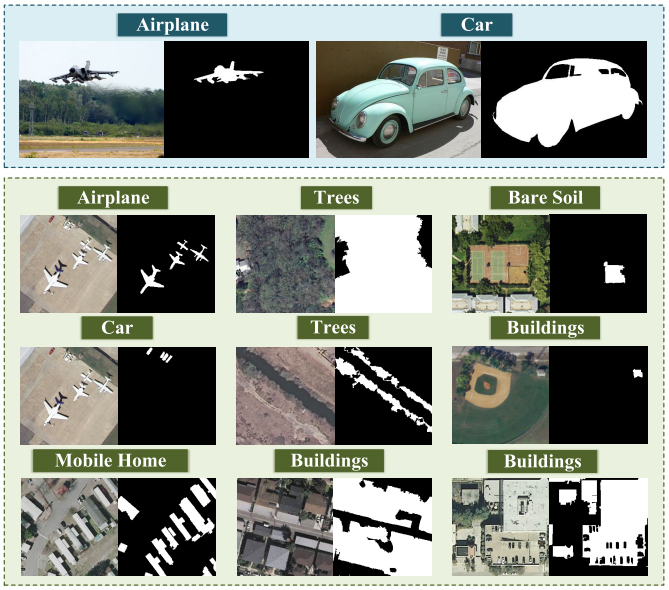


Fig. 1. Some representative cases between the natural image and remote sensing image segmentation. The cases in the blue dotted box are from the PASCAL-5^i dataset, and the cases in the green dotted box are from the DLRSD-4^i dataset we reconstructed.

segmentation algorithms are developed with the natural images. However, there are significant differences between the natural images and remote sensing aerial images. As shown in the comparison of the two categories "airplane" and "car" in Fig. 1, the natural images are generally viewed from a supine or flat perspective to capture the targets, while the aerial images taken by high-altitude equipment such as satellites are generally viewed from a top-down perspective. It tends to lose the original distinctive features of the segmentation target. For example, a car viewed from a flat or elevated perspective can be easily judged based on the overall outline or unique components such as wheels and windows, while the overhead angle of the car is basically rectangular, the iconic features are almost all lost thus increasing the difficulty of the target being segmented.

3) *Large Intra-Class Diversity of Segmentation Targets:* Due to the different altitudes of the satellites and the variety of scenes taken, the same segmentation target in different scenes accounts for a very large difference in the proportion of the image and the style of representation, as given in the case of "trees" and "buildings" from Fig. 1. Because of the diversity of species and the nature of their environment, hundreds of trees exist in different landscapes such as jungles, deserts, and forests, and there is a great deal of variability in their characteristics. Similarly, due to different architectural styles and building scales in different scenes, it is difficult to accurately segment the unlabeled buildings with only a small number of reference samples, especially in the case of large intra-class diversity.
4) *High Similarity Between Different Categories:* One of the challenges of natural image segmentation is the difficulty in distinguishing the target from the irrelevant background, which is also present in aerial images. As in the example of "bare soil" in Fig. 1, the soil is more difficult to distinguish from the unfinished basketball court in both texture and shape. Besides, if classes with similar features are both present in the support set, such as "mobile home" and "buildings" in Fig. 1, it is a difficult challenge to accurately segment the query of these two categories.

These dilemmas put forward more stringent requirements for few-shot aerial image segmentation researches. The currently popular prototype-based correlation studies, [13], [14], [18], [19], although achieving a more ingenious and streamlined network architecture, ignore the interconnectedness of segmentation targets between foreground and background in query and support images [27]. Given the intensive nature of few-shot segmentation task, Min et al. [15], Hong et al. [16], and Peng et al. [17] recently proposed to explore the pixel correlation between query features and foreground support features to avoid information compression in prototyping. However, for the problem of high similarity between background and segmented targets in aerial images, only constructing the correlation between foreground largely leads to the loss of background–foreground potential information [15], [27]. To alleviate the above problems, we incorporate the self-attention mechanism to capture the internal correlation of features at multiple feature scales, constructing a dual correlation of query and support feature foreground–background information. In addition, to enhance the network in capturing global critical information, we combine the coordinate attention in multiscale fusion. Based on the above feature-level and pixel-level designs, our network achieves a great improvement in few-shot segmentation of aerial images, both in terms of capturing key information between global features horizontally and vertically, and in terms of self-supervised aggregation of foreground–background information correlations from multiscale features.

Nevertheless, whether the network is designed at the prototype or pixel level, the corresponding features need to be extracted first. The extraction of representative features depends on the generalization performance of the embedding, while the size of current aerial data is not sufficient to support pre-training a well-informed backbone. In recent years, the rise of self-supervised learning (SSL) [28], [29] is expected to break this bottleneck by defining an pretext task to subtly construct the pseudo-labeled supervised from unlabeled data. However, most of the current SSL work in the remote sensing mainly focus on the classification tasks [30], [31], [32], [33], [34], [35], [36], and few untargeted SSL studies have been designed to address the few-shot aerial image segmentation task. In this article, we combine the advantages of representative SSL frameworks [37], [38], [39], and incorporate the designed asymmetric momentum contrastive learning (AMCL) into the pre-training stage to improve the advanced semantic information representation and generalization ability for the downstream few-shot segmentation task. More importantly, it is encouraging that AMCL can be implemented not only in the pre-training phase of our proposed network, but also in other few-shot segmentation algorithms.

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Furthermore, to enrich the available datasets of few-shot remote sensing aerial imagery, we reconstructed two novel datasets, called DLRSD-4^i and iSAID-4^i, according to the paradigm of few-shot segmentation. Overall, the major contributions of this article can be summarized as follows.

1) We propose a multiscale self-attention aggregation network (MS^{2}A^{2}Net) to improve the few-shot performance of remote sensing aerial image segmentation in terms of data and network architecture. Compared with the prototype-based few-shot segmentation methods, MS^{2}A^{2}Net explores at the pixel-level for the self-aggregation correlation between query and support features.
2) We combine the AMCL in the pre-training phase to improve feature extraction for the downstream few-shot segmentation task. The AMCL does not require any labeled data, but rather improves the discriminability of the encoder for similar class features by constructing the self-supervised auxiliary contrastive learning task.
3) In terms of network architecture, the self-attention aggregation of query and support features is designed at multiscales to explore the similarity of foreground and background information in the pixel-level. Besides, multiscale fusion and coordinate attention mechanism are combined to improve the network's perception of key global features in upsampling.
4) We reconstructed two benchmark datasets for few-shot object segmentation of remote sensing aerial imagery, dubbed DLRSD-4^i and iSAID-4^i. We extensively evaluate the proposed MS^{2}A^{2}Net on three benchmark datasets, and experimental results show that MS^{2}A^{2}Net achieves competitive performance over existing state-of-the-art (SOTA) methods.

The rest of this article is structured as follows. Section II briefly reviews the related work about few-shot learning for semantic segmentation and self-supervised few-shot learning. In Section III, the proposed MS^{2}A^{2}Net is described in detail. Section IV reports the experimental results with comprehensive analysis. Conclusion is drawn in Section V.

II. Related Work

A. Few-Shot Learning for Semantic Segmentation

Semantic segmentation with supervised learning aims to classify each pixel in an image into a set of predefined categories. Few-shot semantic segmentation, on the other hand, generalizes the segmentation ability of the network to any unseen category under the constraint of only a few annotated samples [27].

In recent years, massive excellent few-shot semantic segmentation works have emerged. Tian et al. [14] proposed the prior guided feature enrichment network (PFENet) to alleviate the generalization loss problem caused by the misuse of high-level features through a training-free prior mask generation method. A feature enhancement module is also designed in PFENet to adaptively enhance the query features with support features and prior masks to overcome the spatial inconsistency between query samples and support samples. In [16], the volumetric aggregation with transformers (VATs) is designed to efficiently process high-dimensional correlation graphs between support sets and query sets with both convolutional neural networks (CNNs) and Transformers. Min et al. [15] proposed the hypercorrelation squeeze network (HSNet), which constructs the correlation of multilevel features and the efficient 4-D convolution, and extracts different features from different intermediate convolution layers to construct a 4-D correlation tensor. Lang et al. [13] incorporated an additional base learner into the meta-learner, called BAM, to explicitly identify regions that do not require segmentation. An extended version of BAM is fully studied in [40]. Peng et al. [17] designed a hierarchically decoupled matching network (HDMNet) based on the Transformer architecture by introducing the self-attention module to assist in building hierarchical dense features, while utilizing a matching module and correlation distillation to reduce overfitting and facilitate fine-grained segmentation.

In terms of few-shot semantic segmentation of remote sensing images, it is still in an emerging stage due to the scarcity of datasets and targeted models. Wang et al. [25] proposed a deep meta-metric learning framework called DMML-Net, which consists of a meta-metric learner and a base metric learner. In [26], a novel few-shot aerial image segmentation dataset iSAID-5^i is published. In addition, a matching module and a scale-aware focal loss are designed to provide more accurate parsing guidance on the problem of scale imbalance. Lang et al. [18] designed two components, the progressive parsing module and the commonality distillation module, to compensate for the impact of incomplete targets and interference from other irrelevant targets in the segmentation scene. In [19], the global rectification and decoupled registration are designed to achieve more targeted identification of the positioning of segmentation targets.

B. Self-Supervised Few-Shot Learning

SSL aims to enhance the general feature representation ability of the backbone by constructing pretext tasks without relying on any labeled data to improve the transfer performance on downstream tasks [41]. It has been widely studied in the fields of remote sensing for monocular depth estimation [28], change detection [29], [42], hyperspectral unmixing and classification [30], [31], [32], scene classification [33], [34], [36], object detection [43], [44], [45], etc. The related self-supervised remote sensing semantic segmentation methods [46], [47], [48] have also received widespread attention recently.

In the self-supervised few-shot learning of remote sensing, many excellent studies have been proposed. Self-supervised auxiliary tasks, such as contrastive learning [35], [49], [50], [51], [52], [53], patch location [35], rotation prediction [35], [50], [54], and jigsaw puzzle [35], are deeply explored in remote sensing scene classification. Inkawhich [55] integrated the classic self-supervised contrastive learning framework SimCLR [37] into the target recognition task of synthetic aperture radar (SAR). Xue et al. [56] constructed multitask contrastive learning in land cover classification of multimodal

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remote sensing images. In terms of PolSAR images, relevant self-supervised contrastive tasks are also specially designed for ship detection [57] and classification [58]. In addition, the Bootstrap Your Own Latent (BYOL) [39] combined with self-distillation strategy is studied in remote sensing image captioning [59]. While Monteiro et al. [60] employed three self-supervised auxiliary tasks, including rotation, jigsaw, and frame order prediction, to improve the performance of seismic image segmentation. However, currently in the field of remote sensing, there are relatively few related studies on self-supervised few-shot segmentation, and the related researches mainly focus on natural images [61], medical images [62], and point cloud data [63].

III. Methodology

A. Overview

In this work, we strive to enhance the performance of few-shot aerial image segmentation by reducing the dependence on annotated data and designing a pixel-level network architecture. To obtain a better feature representation in the downstream few-shot segmentation task, we first construct the AMCL as a self-supervised auxiliary task to train the backbone. After the training is completed, except the base encoder is frozen to be used as the feature representation for the downstream few-shot segmentation task, other components including the momentum encoder, projector, and predictor are discarded.

Subsequently, we train the MS^{2}A^{2}Net based on the meta-learning paradigm (also known as episodic training) consistent with most existing few-shot segmentation researches [13], [14], [15], [16], [17]. Without loss of generality, given two datasets $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ that are disjoint on the segmentation targets, i.e., $\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \varnothing$. The network is expected to complete training on the $\mathcal{D}_{\text{train}}$ with a sufficient number of labeled samples and generalize the experience to the unseen categories in $\mathcal{D}_{\text{test}}$, with only a small number of labeled samples as a reference for accurate segmentation of unlabeled images. In the K-shot learning task, both $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ consist of multiple episodes, each constructed from a support set $\mathcal{S} = \{(I_s^k, M_s^k)\}_{k=1}^K$ and a query set $\mathcal{Q} = \{(I_q, M_q)\}$, where $I \in \mathbb{R}^{3\times H\times W}$ represent the raw image and $M \in \{0,1\}^{H\times W}$ is the binary mask. Overall, the MS^{2}A^{2}Net is designed to learn a mapping in multiple iterations of episodes during training, to more accurately predict the query mask M_q based on the support set $\mathcal{S}$ and the query image I_q. Once the training is completed, the parameters of the MS^{2}A^{2}Net are frozen, and the inference is performed on several episodes in $\mathcal{D}_{\text{test}}$ to report the overall performance.

Furthermore, we note that individual components of our framework appear in different instances in previous work. In terms of self-supervised contrastive learning, there are momentum contrast in MoCo [64], projection head in SimCLR [37], prediction head in BYOL [39], and some combination extensions [38], [65], [66]. In terms of network architecture, there are multiscale feature fusion in feature pyramid network [67], attention mechanisms [68], [69], and Vision Transformers [70]. However, it is necessary to emphasize that the superiority of our framework over previous work is not determined by any single component, but by their effective composition and comprehensive design in the few-shot aerial imagery segmentation task. Moreover, it is important to highlight that the overall design of our components and framework is innovative and has not been seen in any previous work on few-shot aerial image segmentation.

B. Asymmetric Momentum Contrastive Learning

The image segmentation task relies heavily on the feature representation capabilities of the backbone [36]. While SSL pre-training has surpassed supervised learning counterpart by constructing auxiliary tasks that do not require labeled mask [38], [39], [66]. Therefore, without relying on any label data, we constructed the SSL to enhance the global and local feature representation of the backbone in the pre-training phase with the introduced AMCL. The overall framework is shown in Fig. 2.

Specifically, an image I is randomly augmented to obtain two views I_1 and I_2. The overall architecture has two asymmetric branches to process these two views. One branch contains the base encoder updated with gradients, the projector and the predictor, while the other branch consists of the momentum encoder and the projector. The base encoder and the momentum encoder are the same architecture but with different update strategies, defined by f_q and f_k, respectively. Without loss of generality, both encoders can be either CNNs or Transformers. Furthermore, the projector ($\mathcal{P}_{\text{ro}}$) and the predictor ($\mathcal{P}_{\text{re}}$) in two branches are composed of two-layer multilayer perceptron (MLP).

Formally, the base encoder outputs a representation $y_{\theta_k} \triangleq f_{\theta_k}(I_1)$, which is then input to the projector and predictor in sequence to obtain the query q, i.e., $z_{\theta_k} \triangleq \mathcal{P}_{\text{ro}}(y_{\theta_k})$ and $q \triangleq \mathcal{P}_{\text{re}}(z_{\theta_k})$. On the other branch, the augmented view I_2 only needs to successively input the momentum encoder and projector to obtain the positive key k^+, i.e., $y_{\theta_q} \triangleq f_{\theta_q}(I_2)$ and $k^+ \triangleq \mathcal{P}_{\text{ro}}(y_{\theta_q})$.

In contrastive learning frameworks such as BYOL [39] and SimSiam [65] that do not require negative examples, the loss function generally directly computes the similarity between the representations of two augmented views. These frameworks achieve remarkable improvements for image classification task that relies on semantic information from global features. However, in the pixel-level classification task of few-shot image segmentation, the comparison of local subtle features is equally important. This requires the base encoder to be as informative as possible in SSL to compare the features of other different categories. Based on this consideration, we adopted the maintenance of dictionaries as data sample queues in MoCo [38], [64]. To be specific, a dynamic dictionary, with a large capacity and a fixed number, is constructed to provide enough negative samples for comparison, but with features that are slowly updated with each mini-batch iteration. The representations of negative examples k^- are likewise obtained by the momentum encoder, i.e., $k^- \triangleq \mathcal{P}_{\text{ro}}(f_{\theta_q}(I_{k^-}))$, where I_{k^-} is the corresponding raw image. In the comparison of the dictionary queue, the aim is to make the query q as similar

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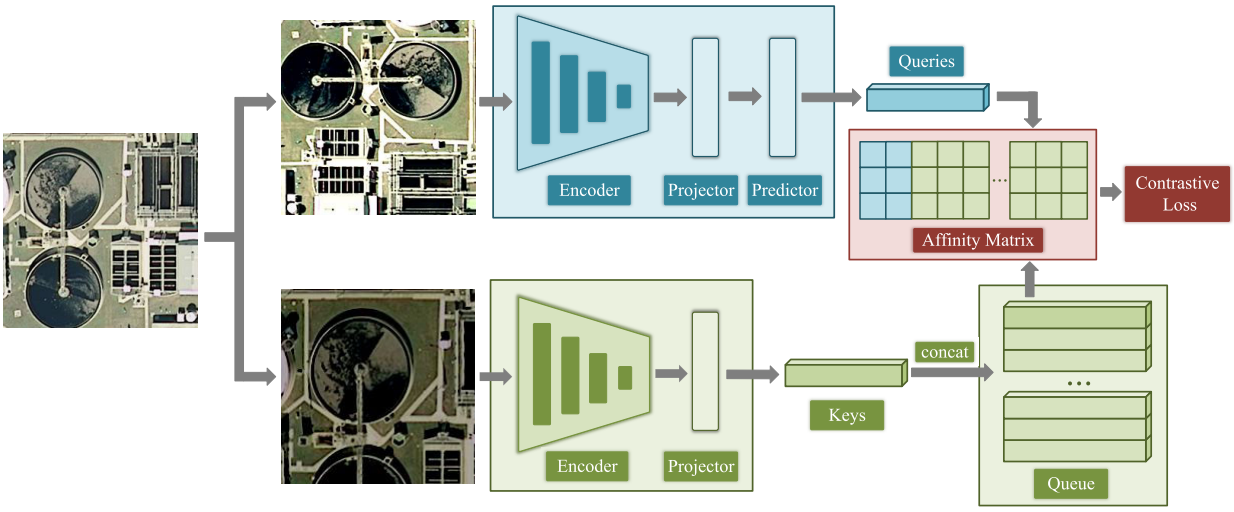


Fig. 2. Overall framework of the AMCL. The k^- in the queue is slowly updated in each iteration of the mini-batch with the momentum encoder. An image I is augmented to obtain two different views I_1 and I_2. I_1 is successively input into the base encoder, the projector, and the predictor to obtain the query q. While I_2 is input into the momentum encoder and the projector to obtain the positive key k^+. Subsequently, k^- is concatenated to the queue to compute the affinity matrix with query q. After the training is completed, only the base encoder is frozen and transferred to the downstream few-shot segmentation task.

as possible to its own k^+ of the same view and away from the unsimilar k^-. Therefore, the employed InfoNCE [71] loss function can be expressed as

$$\mathcal{L}_q = -\log \frac{\exp\left(q \cdot k^+/\tau\right)}{\exp(q \cdot k^+/\tau) + \sum_{k^-} \exp(q \cdot k^-/\tau)} \tag{1}$$

where τ stands for the temperature hyperparameter. The value of τ should not be set too large or too small. If the τ value is set too large, the model treats all negative examples equally and lacks targeted comparative learning. On the contrary, if the value of τ is set too small, the model only focuses on those samples that are particularly difficult to converge, and the learned features will not be easily generalized.

Consequently, the parameters of the base encoder are updated online, while the parameters of the momentum encoder are updated with a momentum strategy to ensure that the examples in the queue are in a relatively consistent feature. Their respective parameters updates are as follows:

$$\theta_q \leftarrow \text{Optimizer}\left(\theta_q, \nabla \mathcal{L}_q, \eta\right) \tag{2}$$

$$\theta_k \leftarrow \gamma\theta_k + (1-\gamma)\theta_q \tag{3}$$

where η is the learning rate and the γ indicates a momentum coefficient with values in the range [0, 1). In particular, to keep the consistency of features in the queue and avoid the model learning some shortcut solutions, the momentum coefficient γ should be set relatively large.

After the training is completed, the momentum encoder, all projectors and predictors are discarded except for the base encoder which is frozen and transferred to the downstream few-shot aerial image segmentation task.

C. Multiscale Self-Attention Aggregation Network

In the existing representative few-shot semantic segmentation studies, most of the work [13], [14] is based on compression into class-level prototypes, which saves computational cost but inevitably results in critical information loss. As a result, the segmentation performance is heavily influenced by the backbone from which the prototypes are extracted. Therefore, there are some related works focusing on the pixel-level, such as hypercorrelation construction [15] and cost aggregation [16], but it is limited to foreground information. While the foreground information can provide fine-grained associated information for the segmented target, ignoring the background information makes it difficult to distinguish objects that are similar to the target to be segmented.

To overcome these, we resort to multiscale self-attention to aggregate the foreground and background pixel-level associations between query and support features. The specific MS²A²Net is shown in Fig. 3. In detail, the features of four different scales of the support and query sets are first obtained with the previously self-supervised pre-trained backbone. The support and query feature maps of the nth layer in the mth scale are denoted by $\mathcal{F}_s^{m,n}$ and $\mathcal{F}_q^{m,n}$, respectively. In terms of the specific values, relative to the input images $I \in \mathbb{R}^{3\times H\times W}$, the four different scales $m \in \{\mathbb{R}^{c_i\times(H\times W)/2^{(i+1)}}\}_{i=1}^4$, which is in line with the stage output of most current CNN and Transformer architectures, and the $n \in \{1, \ldots, N_m\}$ is the index of all intermediate layers in mth scale. Besides, the support masks are downsampled by bilinear interpolation to obtain three different scales of mask respectively, i.e., $M_s^j \in \{\mathbb{R}^{(H\times W)/2^{(j+2)}}\}_{j=1}^3$.

1) Multiscale Self-Attention Aggregation: The support and query features at three different scales combined with the corresponding support masks to perform self-attention aggregation processing. To avoid tedious repetition, we only describe the operation of the last scale in detail, and the other two scales are implemented in the same way. First, it is necessary to perform a Flatten operation to achieve a dimensionality reduction as follows:

$$\mathcal{F}_q \in \mathbb{R}^{c_4\times\frac{HW}{32}} \leftarrow \text{Flatten}\left(\mathcal{F}_q \in \mathbb{R}^{c_4\times\frac{H\times W}{32}}\right) \tag{4}$$

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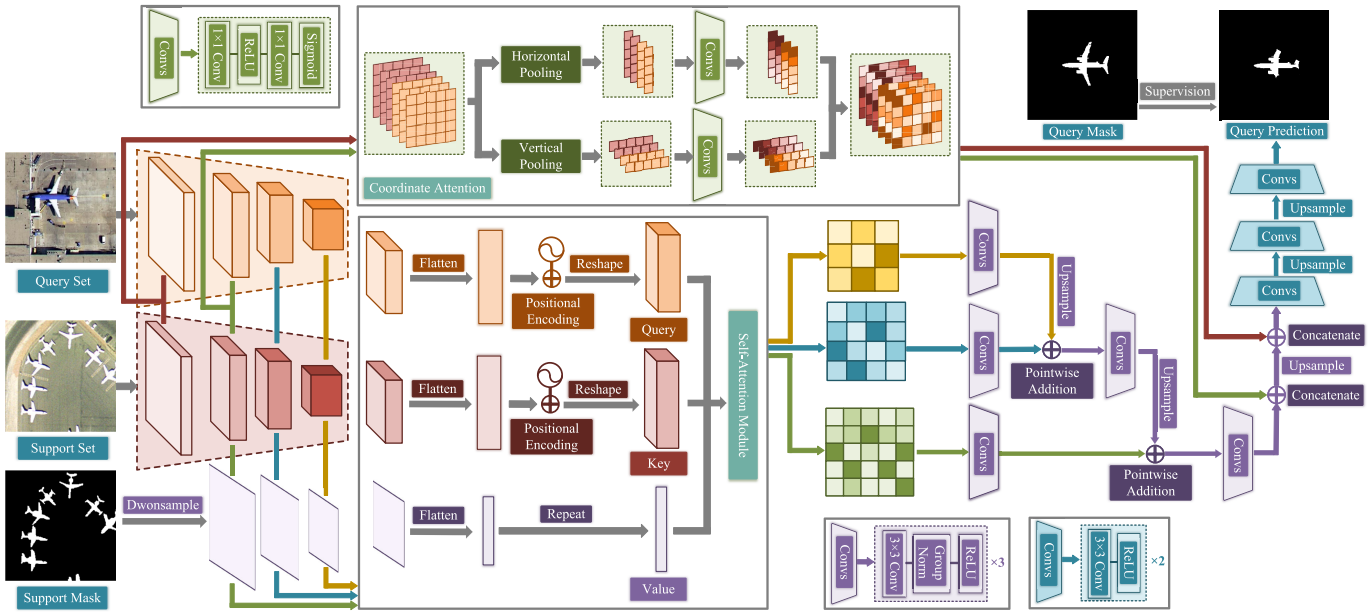


Fig. 3. Overview of the MS^{2}A^{2}Net. We show here the case of 1-shot setting in the training phase. After the training is completed, all the parameters are frozen and inference is performed based on the test requirements by inputting the corresponding support images and their corresponding masks, as well as the query image into the network and outputting the predicted query mask.

$$\mathcal{F}_s \in \mathbb{R}^{c_4 \times \frac{KHW}{32}} \leftarrow \text{Flatten}\left(\mathcal{F}_s \in \mathbb{R}^{c_4 \times \frac{H \times W}{32}}\right) \tag{5}$$

$$M_s \in \mathbb{R}^{1 \times \frac{KHW}{32}} \leftarrow \text{Flatten}\left(M_s \in \mathbb{R}^{1 \times \frac{H \times W}{32}}\right) \tag{6}$$

where K is the number of samples in the support set. This step can be interpreted as treating each pixel as a token. It is worth noting that the variation of data dimensions we give a distinction by "$\times$."

Both $\mathcal{F}_q$ and $\mathcal{F}_s$ then need to be processed by the Positional Encoding and Linear Projections commonly used in the Transformer architecture. Immediately afterward, the features and the mask perform the Reshape operation and the Repeat operation, respectively, with the specific dimensions change as follows:

$$\mathcal{F}_q \in \mathbb{R}^{\frac{c_4}{\mathcal{H}} \times \frac{HW}{32} \times \mathcal{H}} \leftarrow \text{Reshape}\left(\mathcal{F}_q \in \mathbb{R}^{c_4 \times \frac{HW}{32}}\right) \tag{7}$$

$$\mathcal{F}_s \in \mathbb{R}^{\frac{c_4}{\mathcal{H}} \times \frac{KHW}{32} \times \mathcal{H}} \leftarrow \text{Reshape}\left(\mathcal{F}_s \in \mathbb{R}^{c_4 \times \frac{KHW}{32}}\right) \tag{8}$$

$$M_s \in \mathbb{R}^{1 \times \frac{KHW}{32} \times \mathcal{H}} \leftarrow \text{Repeat}\left(M_s \in \mathbb{R}^{1 \times \frac{KHW}{32}}\right) \tag{9}$$

where $\mathcal{H}$ is the number of heads in multihead attention of Transformer. Then, $\mathcal{F}_q$, $\mathcal{F}_s$, and M_s are input into the self-attention module as Query (Q), Key (K) and Value (V), respectively. The output self-attention aggregated feature $\mathcal{F}_{\text{SA}}$ is as follows:

$$\mathcal{F}_{\text{SA}} \in \mathbb{R}^{1 \times \frac{H \times W}{32}} \leftarrow \text{Attention}\left(\mathcal{F}_q, \mathcal{F}_s, M_s\right). \tag{10}$$

Intuitively, for a particular query pixel, $\mathcal{F}_q \mathcal{F}_s{}^T$ is executed to obtain its similarity to all supported pixels. It is then multiplied with M_s to aggregate the support masks and weighted by similarity. Based on this, the foreground and background pixels of the query features in the weighted aggregation are focused to correspond to the foreground and background pixels of the support features for effective metric learning. Finally, a total of three different scales of self-attention aggregation features can be obtained, i.e., $\mathcal{F}^j_{\text{SA}} \in \{\mathbb{R}^{1 \times (H \times W)/2^{(j+2)}}\}_{j=1}^{3}$. By constructing the self-attention aggregated associations of query and support features at three different scales, finer-grained multiscale pixel-based information can be effectively extracted while exploring the similarity not only in foreground information but also expanding to the combination of background information.

2) Multiscale Feature Fusion: Subsequently, these three features at different scales increase their number of channels by three repeated convolution blocks. Each convolutional block consists of a 3 × 3 convolutional layer, Group Normalization, and ReLU activation function. $\mathcal{F}^3_{\text{SA}}$ took the lead in using bilinear interpolation to upsample and combine with $\mathcal{F}^2_{\text{SA}}$ by the Pointwise Addition, followed by three convolution blocks to increase the number of channels, and performs an upsampling operation and Pointwise Addition with $\mathcal{F}^1_{\text{SA}}$.

After the features of these three scales are fused, the number of channels is increased by three convolution blocks, and a skip-connection is concatenated with $\mathcal{F}^2_{\text{CA}}$, where $\mathcal{F}^2_{\text{CA}}$ is the features of $\mathcal{F}^2_q \oplus \mathcal{F}^2_s$ via the coordinate-attention path. The coordinate-attention path is designed mainly to enhance perceptual coarse global information to compensate for the absence of pixel-based to the global key features. First, the query and support features in the same scale are concatenated, i.e., $\mathcal{F}^2 \leftarrow \mathcal{F}^2_q \oplus \mathcal{F}^2_s$. Second, the average pooling of the features along the vertical and horizontal directions in each channel c of the feature $\mathcal{F}^2$ is calculated as follows:

$$\mathcal{F}^{2,c}_X(x) \leftarrow \frac{1}{|Y|} \sum_{z=1}^{|Y|} \mathcal{F}^{2,c}(x, z) \tag{11}$$

$$\mathcal{F}^{2,c}_Y(y) \leftarrow \frac{1}{|X|} \sum_{z=1}^{|X|} \mathcal{F}^{2,c}(z, y) \tag{12}$$

where $\mathcal{F}^{2,c}_X(x)$ and $\mathcal{F}^{2,c}_Y(y)$ represent the average aggregation of all feature vectors of column x and row y into one

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feature vector, respectively. $|X|$ and $|Y|$ denote the number of horizontal and vertical feature vectors, respectively.

Subsequently, $\mathcal{F}_X^{2,c}(x)$ and $\mathcal{F}_Y^{2,c}(y)$ are squeezed by a 1×1 convolutional layer, and the number of channels is recovered by another 1×1 convolutional layer after the ReLU activation function. The obtained horizontal and vertical features are input to the Sigmoid layer and the weights of each direction are output as follows:

$$\mathcal{F}_X'^{2,c}(x) \leftarrow \text{Sigmoid}\left(\text{Convs}\left(\mathcal{F}_X^{2,c}(x)\right)\right) \tag{13}$$

$$\mathcal{F}_Y'^{2,c}(y) \leftarrow \text{Sigmoid}\left(\text{Convs}\left(\mathcal{F}_X^{2,c}(x)\right)\right). \tag{14}$$

Based on this, the weights of the two spatial coordinate directions are fused by matrix multiplication to obtain a unique weight for each position as follows:

$$\mathcal{F}_{\text{CA}}^2 \leftarrow \mathcal{F}_X'^{2,c}(x) \times \mathcal{F}_Y'^{2,c}(y). \tag{15}$$

After the $\mathcal{F}_{\text{CA}}^2$ and the fusion features are concatenated, upsampling is performed to raise the feature size to $(H \times W)/8$, and a skip-connection is concatenated with $\mathcal{F}_{\text{CA}}^1$. $\mathcal{F}_{\text{CA}}^1$ is obtained in the same way as (11)–(14), except that the input features are $\mathcal{F}_q^1 \oplus \mathcal{F}_s^1$.

There are two major differences in dimension between the current fused features and the query mask $M_q \in \{0, 1\}^{H \times W}$ that needs to be predicted, namely an excessive number of channels and a narrow scale. Therefore, we need to reduce the number of channels of fused features by three consecutive convolution blocks and interpolate into the upsampling operation. Each convolution block consists of a 3×3 convolutional layer and a ReLU activation function, which is different from the previous convolutional blocks. Finally, the network outputs the predicted query mask $\hat{M}_q$.

In the training phase, the binary cross-entropy loss function is employed to update the network parameters

$$\mathcal{L}_{\text{BCE}} = \frac{1}{H \times W} \sum_{l=1}^{H \times W} M_q^l \log \hat{M}_q^l + \left(1 - M_q^l\right) \log\left(1 - \hat{M}_q^l\right). \tag{16}$$

After training is completed, all parameters of the entire MS²A²Net including the backbone and the convolutional layers connected behind are frozen. In the inference implementation stage, according to the K-shot segmentation setting, K support images and the corresponding masks are input to the network, combined with the query image, the MS²A²Net finally outputs the predicted query mask.

IV. Experimental Studies

A. Datasets

1) DLRSD-4^i: We reconstructed a novel few-shot remote sensing aerial image segmentation dataset, dubbed DLRSD-4^i, from the DLRSD dataset [22], [23] according to the paradigm of few-shot segmentation. Table I gives the detailed segmentation targets for each fold in the DLRSD-4^i dataset. In DLRSD dataset [22], [23], there are 17 segmentation targets in total, but considering that the cases in the category "filed" are basically all foreground information without reference value for the background information, so it is discarded in DLRSD-4^i, leaving only the following 16 segmentation targets. This dataset has a total of 7002 images with 256×256 pixels. These 16 segmented object classes in the DLRSD-4^i dataset are equally divided into four folds, 3 of which are used for training and 1 for testing. The DLRSD-4^i dataset is small in scale and the scene differences are large, which puts forward strict requirements on the fast generalization ability of the few-shot segmentation algorithms.

TABLE I
SPECIFIC INFORMATION OF EACH SPLIT IN THE DLRSD-4^i DATASET

DLRSD-4^i	Categories	Images
DLRSD-4^0	Airplane; Bare Soil; Cars; Chaparral	1855
DLRSD-4^1	Buildings; Court; Dock; Grass	1894
DLRSD-4^2	Mobile Home; Pavement; Sea; Sand	1824
DLRSD-4^3	Ship; Tanks; Trees; Water	1429

2) iSAID-5^i [26]: Yao et al. [26] reintegrated the iSAID dataset [24] according to the commonly used few-shot semantic segmentation paradigm and released the iSAID-5^i dataset. iSAID-5^i dataset is the only publicly available benchmark dataset for few-shot remote sensing segmentation and has been extensively studied in [18], [19], and [26]. This dataset contains 2806 high-resolution aerial images, including 6 55 451 object instances from 15 categories.

3) iSAID-4^i: To enrich the diversity of few-shot remote sensing image segmentation datasets, especially taking four categories as a split option, based on the iSAID-5^i [26] dataset, we added the category of buildings from the Massachusetts dataset [72], and reconstructed a novel few-shot remote sensing segmentation dataset named iSAID-4^i. On the one hand, buildings in aerial imagery are important for applications such as urban change monitoring and urban planning, so it is valuable to be considered for inclusion in few-shot learning [25]. On the other hand, the generalization performance of few-shot segmentation is an important measure of the model, and keeping the same number of folds of different datasets enables generalization testing without model fine-tuning. The specific information of each split in the iSAID-4^i dataset is shown in Table II. The iSAID-4^i dataset has 28 214 images with a size of 256×256 pixels. Specifically, the iSAID-4^i dataset has a total of 16 categories of segmentation targets, and the cross-validation strategy is implemented, i.e., four categories are used for testing and the other 12 categories are used for training. The large size and rich scenes of the iSAID-4^i dataset impose stringent requirements on few-shot segmentation algorithms to deal with high inter-class similarity and large intra-class variability.

B. Implementation Details

To ensure the fairness and comparability of experiments, we implemented all the experiments on the Pytorch framework with CUDA 11.6 by using a single NVIDIA RTX 3090. The stochastic gradient descent (SGD) is employed as the optimizer for 200 epochs, with the training batch size to 8.

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TABLE II
SPECIFIC INFORMATION OF EACH SPLIT IN THE iSAID-4^i DATASET

iSAID-4^i	Categories	Images
iSAID-4^0	Helicopter; Small Vehicle; Tank; Tennis Court	11850
iSAID-4^1	Basketball Court; Bridge; Large Vehicle; Soccer Field	9175
iSAID-4^2	Field; Ship; Swimming Pool; Roundabout	8551
iSAID-4^3	Buildings; Diamond; Harbor; Plane	9260

The initial learning rate is set to 0.01, with the momentum of 0.9 and the weight decay rate of 10^{-5}. The overall results are reported by 2000 testing episodes. The source code is available. The reconstructed two datasets (DLRSD-4^i and iSAID-4^i) and the code of MS^{2}A^{2}Net are available on https://github.com/OMEGAFSL/MS2A2Net.

In terms of backbones, to verify the universality of our proposed MS^{2}A^{2}Net, three representative few-shot semantic segmentation networks, i.e., ResNet-50, ResNet-101 and Swin-B, are employed to perform the experiments. To ensure a fair comparison, all methods are implemented from their open-source code and reported with their best results. The datasets pre-trained with the proposed AMCL are the two remote sensing segmentation datasets, i.e., DLRSD and iSAID.

In terms of metrics, we report all segmentation results in mean intersection over union (mIoU) and foreground–background IoU (FB-IoU). The mIoU metric refers to the average over the IoU values of all classes in a fold, while FB-IoU ignores the segmented objects and directly calculates the average value of foreground and background IoU. Theoretically, mIoU better reflects the generalization ability and prediction quality of the model than FB-IoU, so mIoU is the metric that should be focused on.

C. Comparison With SOTA

To demonstrate the effectiveness and superiority of our proposed MS^{2}A^{2}Net, for the two reconstructed datasets DLRSD-4^i and iSAID-4^i, five SOTA few-shot semantic segmentation methods are employed as comparison counterparts, namely PFENet [14] (2020 TPAMI), VAT [16] (2022 ECCV), HSNet [15] (2021 ICCV), BAM (2022 CVPR) [13], and HDMNet (2023 CVPR) [17]. Except that PFENet and BAM are prototype-level, other algorithms are the pixel-level in line with our MS^{2}A^{2}Net. The results of all methods are reproduced according to the corresponding papers and published source code, and fine-tuned to obtain their best performance. For the benchmark dataset iSAID-5^i, we follow the few-shot aerial image segmentation work [18], [19], [25] and report based on their comparative experimental results.

1) Experiments on the DLRSD-4^i Dataset: Table III gives a comparison of the MS^{2}A^{2}Net with the five SOTA on the DLRSD-4^i dataset. It can be observed from the table that MS^{2}A^{2}Net ranks first in the mIoU metric, but does not perform as well as other algorithms in FB-IoU. It is worth noting that the mIoU metric deserves more attention than FB-IoU in the field of few-shot segmentation, so our subsequent discussion focuses on the mIoU metric. In terms of overall performance, the pixel-level few-shot segmentation algorithms mostly outperform the prototype-level approaches. For MS^{2}A^{2}Net, the deeper the backbone, the better the segmentation effect, especially for the 1-shot setting. More specifically, it outperforms the second best algorithm (VAT) by only 0.01% with the ResNet-50 and 4.86% with the ResNet-101. The performance improvement exhibited by MS^{2}A^{2}Net is even greater as the number of images in the support set increases. The 1-shot to 5-shot improves 4.92% and 4.72% on the ResNet-50 and ResNet-101 backbones, respectively.

In addition, we also provide the qualitative results of MS^{2}A^{2}Net in 1-shot and 5-shot settings as shown in Fig. 4. It is worth noting that since the training paradigm of few-shot segmentation is episode-based, and the support set in each episode is randomly selected, we are consistent with our peers [13], [14], [15], [16], [17] and only show the visual results of the proposed method. It can be observed from the figure that MS^{2}A^{2}Net achieves a good effect on the rough outline of the segmented target with only one label image as a reference, whether for the segmented target with larger target, such as water, or for the smaller segmented target, such as tanks. Moreover, with the increase of images in the support set, MS^{2}A^{2}Net is more accurate at the edge details of the segmented targets. Nonetheless, for dense and small segmentation targets, MS^{2}A^{2}Net does not process the pixels connected between them ideally. More interestingly, for some inaccurate ground truth, such as the case of the tank with its shaded parts marked in, MS^{2}A^{2}Net can effectively distinguish the real tank from its shaded interference.

2) Experiments on the iSAID-4^i Dataset: The overall performance comparison on the iSAID-4^i dataset is given in Table IV. As can be observed from the table, MS^{2}A^{2}Net also performs well on the large-scale iSAID-4^i dataset, winning the first place under both 1-shot and 5-shot settings. In detail, on 1-shot our method is 0.25% and 0.17% ahead of the second place on ResNet-50 and ResNet-101, respectively. Similarly, the lead of MS^{2}A^{2}Net has been further expanded in 5-shot. Besides, Fig. 5 also shows some representative visualization cases of MS^{2}A^{2}Net, all of which are selected from the results based on the ResNet-101 backbone. For those larger and more regularly shaped targets, such as oil tanks and bridges, MS^{2}A^{2}Net is able to segment the vast majority of the targets accurately. In the 1-shot case, for smaller targets, such as vehicles, MS^{2}A^{2}Net is equally capable of segmenting most of the vehicles with only a limited reference. For some cases where segmentation is difficult, such as the planes in 5-shot, the reference images in the support set contain aircraft with different views and scales, while MS^{2}A^{2}Net can segment the majority of the target by integrating the commonalities of features.

3) Experiments on the iSAID-5^i Dataset: Table V shows the overall performance of the benchmark iSAID-5^i dataset. To ensure the fairness of the comparison, the backbone of each method was uniformly selected as ResNet-50. It can be summarized from the experimental results that MS^{2}A^{2}Net is inferior to the SOTA (R^{2}Net). This result actually maintains similarity with our previous analysis, where the high

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TABLE III

OVERALL PERFORMANCE (IN %) COMPARISON ON THE DLRSD-4^i DATASET, THE BEST RESULT IS INDICATED IN BOLD

Method	Backbone	1-shot						5-shot					
		Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**
PFENet [14]	ResNet-50	7.13	17.24	19.61	14.93	14.73	34.72	7.45	19.31	21.52	16.89	16.29	34.36
VAT [16]		22.60	27.15	34.04	33.59	29.34	**43.32**	25.87	30.32	36.13	37.83	32.54	**44.80**
HSNet [15]		20.77	33.68	29.63	29.04	28.28	37.41	23.54	37.49	34.40	34.02	32.36	40.78
BAM [13]		9.97	18.10	35.43	19.64	20.79	37.11	19.70	24.72	32.81	34.92	28.04	41.53
HDMNet [17]		18.27	16.87	32.85	23.46	22.86	32.52	18.98	26.85	44.01	32.84	30.67	40.68
MS^{2}A^{2}Net		22.60	26.49	41.52	26.80	29.35	39.41	28.23	31.46	46.09	31.31	34.27	44.57
PFENet [14]	ResNet-101	9.25	16.19	30.25	18.54	18.56	36.95	9.83	16.43	30.66	22.91	19.96	37.05
VAT [16]		22.31	26.73	33.94	32.38	28.84	43.62	26.64	29.80	39.09	39.51	33.76	45.15
HSNet [15]		20.32	33.61	42.84	28.61	31.35	41.94	27.39	37.73	49.21	33.60	36.98	49.23
BAM [13]		13.89	23.09	31.49	28.42	24.22	40.01	13.66	28.03	35.06	32.53	27.32	42.24
HDMNet [17]		18.03	11.79	34.81	26.23	22.72	37.11	18.64	18.45	43.27	38.98	29.84	46.19
MS^{2}A^{2}Net		26.29	30.06	40.42	41.33	**34.53**	**48.48**	28.86	34.37	44.79	48.96	**39.25**	**50.90**

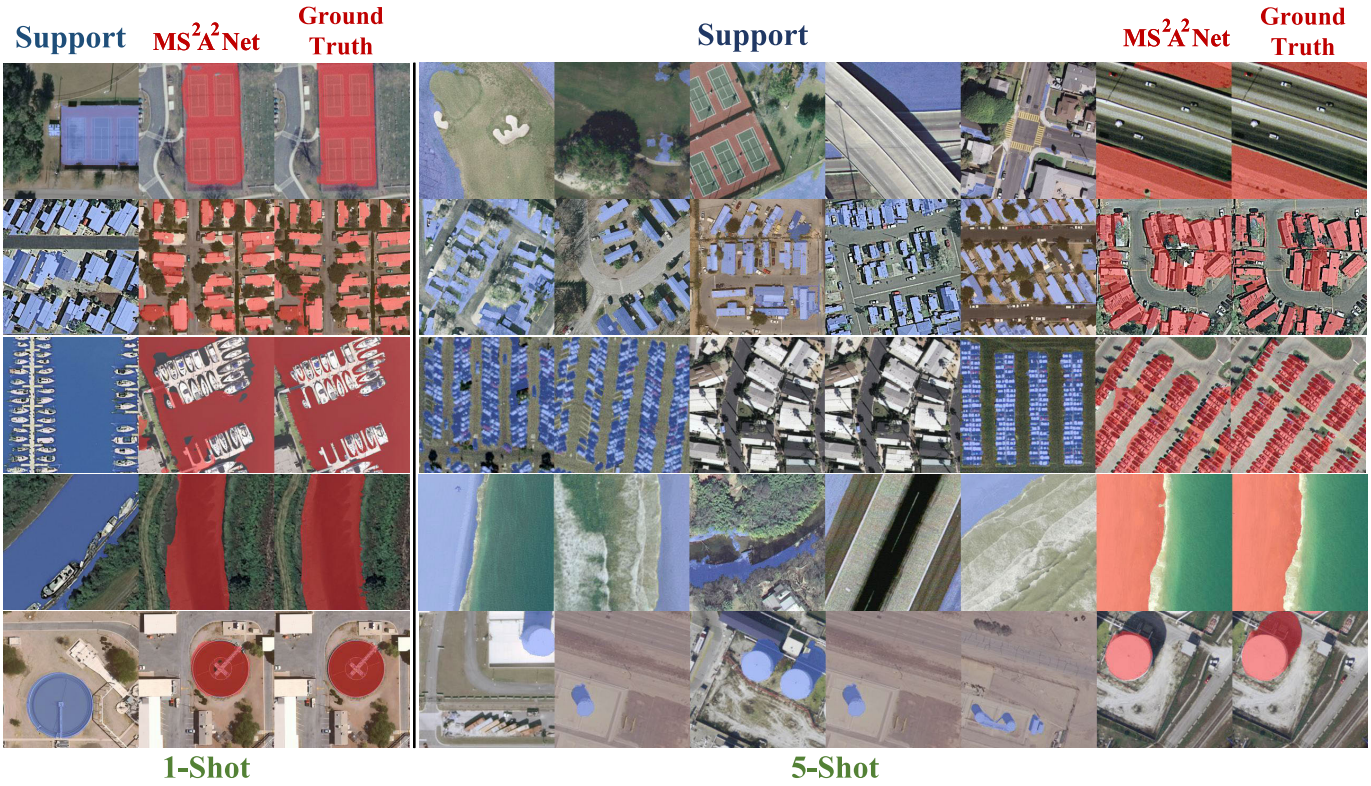


Fig. 4. Qualitative results of the proposed MS^{2}A^{2}Net under 1-shot and 5-shot setting in the DLRSD-4^i dataset. These cases are selected from the ResNet-101 as the backbone.

performance of MS^{2}A^{2}Net actually relies heavily on a large-scale dataset and a deeper feature extraction.

4) Experimental Analysis: In general, for small-scale DLRSD-4^i datasets, MS^{2}A^{2}Net can achieve a more desirable performance with little difference between the support set annotation and query ground truth, but MS^{2}A^{2}Net still needs to be improved for segmentation targets that are smaller or more discrepant. While for large-scale iSAID-4^i dataset, MS^{2}A^{2}Net still has the ability to achieve ideal results with a small number of reference samples of different sizes, even if the segmentation targets of the samples in the support set are significantly different. Meanwhile, the performance of MS^{2}A^{2}Net depends on a large-scale data and a deeper feature extractor for AMCL and the pixel-level self-attention aggregation.

Furthermore, we also implemented MS^{2}A^{2}Net on the currently popular Transformer architecture, and Table VI shows the results on Swin-B in three benchmark datasets. Analyzing the results, the MS^{2}A^{2}Net equipped with Swin-B gives significantly better results on the two benchmark datasets than those equipped with ResNet-50 and ResNet-101. The multiscale self-attention aggregation strategy is closely related to the feature extraction capability of the backbone, so we can improve the performance of the backbone through self-supervised

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TABLE IV

OVERALL PERFORMANCE (IN %) COMPARISON ON THE iSAID-4^i DATASET, THE BEST RESULT IS INDICATED IN BOLD

Method	Backbone	1-shot						5-shot					
		Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**
PFENet [14]	ResNet-50	17.57	25.73	26.45	28.69	24.61	45.96	18.69	27.55	30.46	29.67	26.59	47.98
VAT [16]		34.49	27.18	22.42	19.30	25.85	51.48	37.94	31.38	31.43	19.99	30.18	52.73
HSNet [15]		31.10	22.36	24.83	30.31	27.15	51.92	33.31	23.18	26.49	31.32	28.58	52.12
BAM [13]		31.58	31.02	23.26	28.43	28.57	52.22	33.56	34.48	31.59	30.94	32.64	55.67
HDMNet [17]		25.23	32.23	24.57	22.13	26.04	51.41	34.89	37.72	35.26	28.21	34.02	53.47
MS^{2}A^{2}Net		30.71	31.77	27.26	25.54	**28.82**	**52.97**	35.02	40.58	37.66	30.20	**35.86**	**57.80**
PFENet [14]	ResNet-101	12.96	19.37	16.89	22.45	19.72	49.99	15.65	22.94	20.49	24.94	21.01	51.22
VAT [16]		34.99	34.43	25.83	28.79	31.01	55.22	37.07	37.85	30.86	31.87	34.41	57.23
HSNet [15]		29.86	31.00	28.30	26.29	28.86	**56.21**	41.57	28.22	29.25	33.35	33.17	57.07
BAM [13]		30.52	27.67	25.79	32.30	29.07	54.39	30.82	32.83	32.43	36.04	33.03	56.46
HDMNet [17]		25.37	30.06	25.33	26.87	26.91	51.63	31.72	33.13	33.56	29.89	32.07	58.46
MS^{2}A^{2}Net		38.43	25.63	28.53	32.12	**31.18**	55.62	34.57	39.14	38.93	31.13	**35.94**	**58.48**

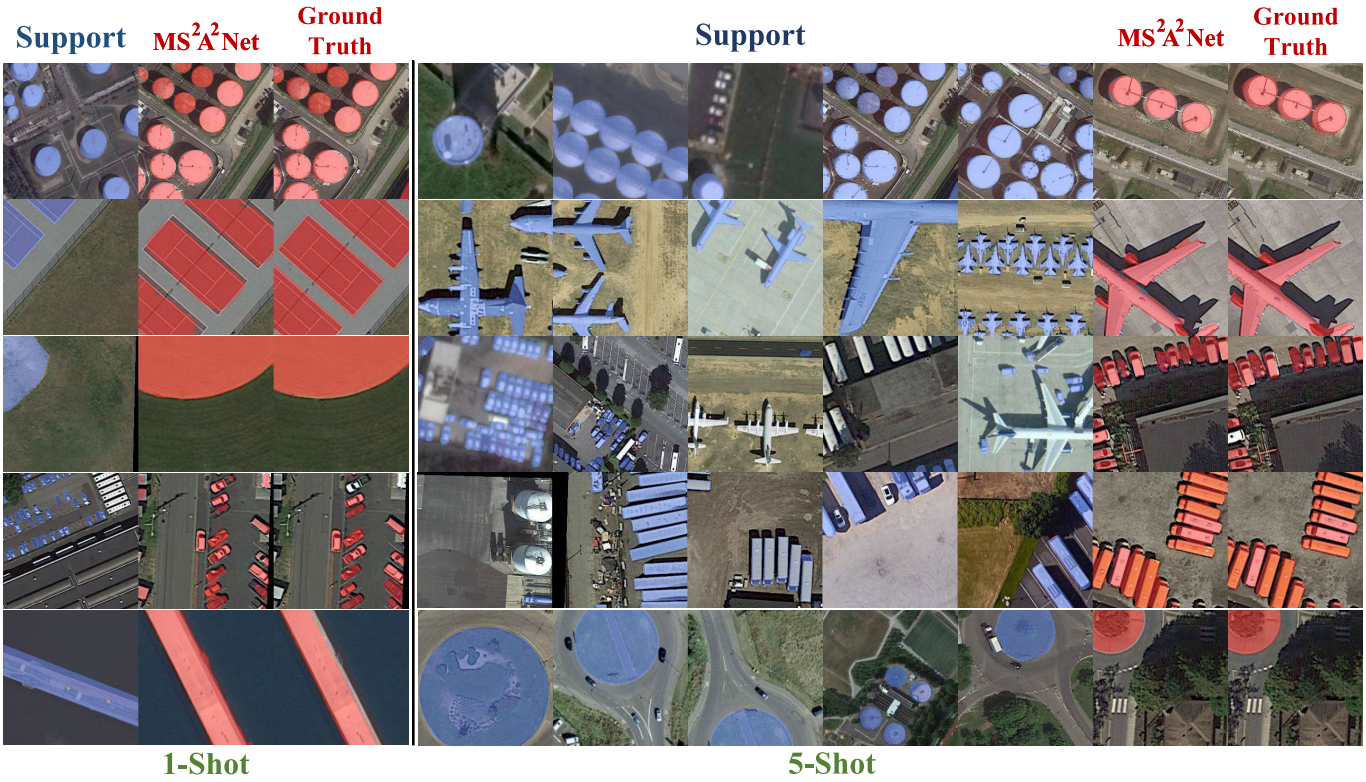


Fig. 5. Qualitative results of the proposed MS^{2}A^{2}Net under 1-shot and 5-shot setting in the iSAID-4^i dataset. These cases are selected from the ResNet-101 as the backbone.

pre-training on the one hand, and deploy it on a more powerful Transformer architecture on the other hand. Nevertheless, this does not mean that our method is without drawbacks. MS^{2}A^{2}Net can easily fail for the segmentation targets that are highly similar to the background or targets with too little annotation in the support set. Some representative failure cases of MS^{2}A^{2}Net on Swin-B backbone are given in Fig. 6.

Although the current SOTA have achieved certain achievements, few-shot aerial image segmentation is still a very difficult problem worthy of further exploration. Due to the fact that most of the objects are rectangular in overlooking angle making the unique features of the target lost, enhancing the discriminative performance of the model from the monotonous features is the focus of the models. Furthermore, the inability to efficiently overcome the large inter-class differences and the intra-class similarities is also the bottleneck of the few-shot aerial image segmentation models.

D. Ablation Studies

1) Ablation Studies on the AMCL: The comparison results of random initialization on the two benchmark datasets, supervised pre-training with ImageNet-1K and pre-training with AMCL, are shown in Fig. 7. The experimental results

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TABLE V

OVERALL PERFORMANCE (IN %) COMPARISON ON THE iSAID-5^i DATASET, THE BEST RESULT IS INDICATED IN BOLD

Method	Backbone	1-shot					5-shot				
		Fold-0	Fold-1	Fold-2	**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2	**mIoU**	**FB-IoU**
PANet [12]	ResNet-50	27.56	17.23	24.60	23.13	56.56	36.54	16.05	26.22	26.27	57.37
CANet [8]		25.51	13.50	24.45	21.15	56.64	29.32	21.85	26.91	26.03	59.46
SCL [10]		34.78	22.77	31.20	29.58	61.30	41.29	25.73	37.70	34.91	64.13
PFENet [14]		35.84	23.35	27.20	28.80	60.09	42.42	25.34	33.00	33.59	63.25
NERTNet [6]		34.93	23.95	28.56	29.15	59.97	44.83	26.73	37.19	36.25	64.45
DCP [11]		37.83	22.86	28.92	29.87	62.36	41.52	28.18	33.43	34.38	63.77
BAM [13]		39.43	21.69	28.64	29.92	62.04	43.29	27.92	38.62	36.61	65.00
DMML-Net [25]		28.45	21.02	23.46	24.31	57.78	30.61	23.85	24.08	26.18	58.26
SDM [26]		27.96	21.99	27.82	25.92	59.58	28.50	25.23	31.07	28.27	59.90
DML [73]		32.96	18.98	26.27	26.07	58.93	33.58	22.05	29.77	28.47	59.23
TBPN [74]		29.33	16.84	25.47	23.88	57.34	30.98	20.42	28.07	26.49	29.67
PCNet [18]		40.24	24.64	31.31	32.06	–	45.31	28.19	37.36	36.95	–
R^{2}Net [19]		41.22	21.64	35.28	**32.71**	**63.86**	46.45	25.80	39.84	**37.36**	**66.18**
MS^{2}A^{2}Net		34.96	25.30	30.71	30.32	59.81	42.30	27.10	36.31	35.24	62.01

TABLE VI

OVERALL PERFORMANCE (IN %) OF MS^{2}A^{2}NET WITH SWIN-B, THE BEST RESULT IS INDICATED IN BOLD

Method	Dataset	1-shot						5-shot					
		Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**
MS^{2}A^{2}Net	DLRSD-4^i	17.01	34.78	44.44	39.53	**33.94**	**42.78**	19.88	42.65	54.65	48.32	**41.38**	**51.12**
	iSAID-4^i	45.35	35.02	34.25	35.60	**37.56**	**59.76**	50.69	42.49	50.81	42.30	**46.57**	**64.06**
	iSAID-5^i	Fold-0	Fold-1	Fold-2		**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2		**mIoU**	**FB-IoU**
		35.32	26.76	32.53		**31.54**	**61.44**	38.92	31.75	40.79		**37.15**	**64.90**

TABLE VII

ABLATION EXPERIMENTS OF MULTISCALE SELF-ATTENTION AGGREGATION ON THE DLRSD-4^i DATASET, THE "✓" INDICATES THAT THE CORRESPONDING SCALE FEATURES ARE AGGREGATED BY SELF-ATTENTION

Scale			1-shot						5-shot					
1/8	1/16	1/32	Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**
✓			16.49	17.41	34.81	18.29	21.75	38.35	18.05	20.39	40.64	19.06	24.54	40.09
	✓		17.50	30.81	44.02	35.20	31.88	42.98	21.46	38.02	55.02	50.25	41.19	52.10
		✓	6.04	13.12	19.44	3.52	10.53	34.42	7.69	13.99	22.60	15.51	14.95	34.78
✓	✓		17.54	35.71	43.66	39.20	34.03	42.82	24.27	42.14	53.23	53.86	**43.38**	**52.43**
✓		✓	14.27	20.64	36.21	21.10	23.06	38.39	14.58	24.87	40.11	23.06	25.66	39.27
	✓	✓	18.61	33.30	46.21	38.04	**34.04**	**42.84**	19.71	41.37	56.62	49.09	41.70	52.33
✓	✓	✓	17.01	34.78	44.44	39.53	33.94	42.78	19.88	42.65	54.65	48.32	41.38	51.12

illustrate that the pre-trained models all outperform the random initialization, but the supervised pre-training with ImageNet-1K is superior. However, it is worth noting that the models pre-trained with ImageNet-1K requires powerful computing power and all labeled data, which is very expensive both in terms of computing resources and sample labeling. In contrast, pre-training with AMCL has very low hardware requirements, and more importantly, it does not require labels and additional data to obtain a well-initialized model. Therefore, considering the comprehensive factors, AMCL takes into account the calculation cost and labeling cost to achieve a compromise between resources and performance.

To further demonstrate the effectiveness of our proposed AMCL, we present the heat map comparison with the randomly initialized model in the iSAID-4^i dataset in Fig. 8. For some targets with a small proportion in aerial images, such as the cases in the second and third rows in the figure, the network pre-trained with AMCL can obviously more easily focus on the key clues most relevant to image target segmentation.

2) Ablation Studies on the Multiscale Self-Attention Aggregation and Coordinate-Attention: Tables VII and VIII show the results of ablation experiments for different scales of self-attention aggregation in the DLRSD-4^i and iSAID-4^i datasets, respectively. The backbone for ablation study is Swin-B. It can

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TABLE VIII

ABLATION EXPERIMENTS OF MULTISCALE SELF-ATTENTION AGGREGATION ON THE iSAID-4^i DATASET, THE "✓" INDICATES THAT THE CORRESPONDING SCALE FEATURES ARE AGGREGATED BY SELF-ATTENTION

Scale			1-shot						5-shot					
1/8	1/16	1/32	Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**	Fold-0	Fold-1	Fold-2	Fold-3	**mIoU**	**FB-IoU**
✓			22.56	20.64	17.66	22.39	20.81	47.87	24.66	25.59	19.57	22.52	23.09	48.82
	✓		46.76	32.43	33.05	34.36	36.65	58.08	49.91	47.21	45.67	42.71	46.38	**65.87**
		✓	25.48	18.44	16.73	26.27	21.73	47.75	29.04	21.96	23.41	26.89	25.33	50.11
✓	✓		47.94	33.12	33.50	35 53	37.52	58.94	50.73	44.37	45.44	43.24	45.95	64.50
✓		✓	27.73	25.58	24.18	26.11	25.90	50.06	31.53	33.58	28.83	30.04	31.00	53.13
	✓	✓	47.26	32.79	35.30	37.01	**38.09**	58.79	52.11	39.79	48.39	44.43	46.18	63.75
✓	✓	✓	45.35	35.02	34.25	35.60	37.56	**59.76**	50.69	42.49	50.81	42.30	**46.57**	64.06

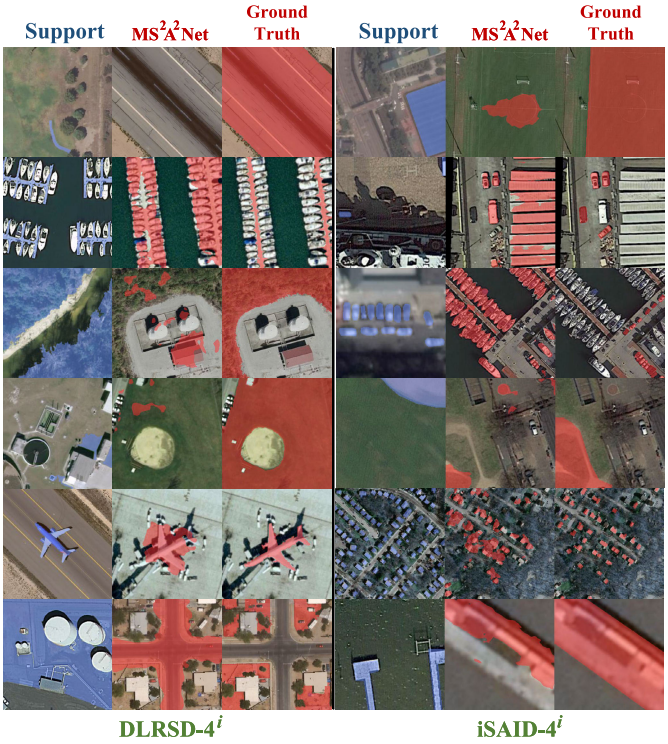


Fig. 6. Some failure cases of MS^{2}A^{2}Net in the 1-shot setting. These cases are selected from the Swin-B as the backbone.

be observed from the results that the self-attention aggregation of (1/16) scale features is crucial, and works much better than the (1/32) scale features used in the prototype-based algorithms. While the aggregation of multiscale, either two-by-two fusion or all fusion, has a certain improvement compared to using only one scale. Interestingly, this does not mean that self-attention aggregation of all features can achieve the best results, which can be reflected in both DLRSD-4^i and iSAID-4^i datasets. More pertinently, MS^{2}A^{2}Net combines the ability of shallow features to capture fine-grained information and deep features to obtain semantic information.

Besides, the enhancement of coordinate-attention path for each fold on the two benchmark datasets is shown in detail in Fig. 9. The lack of pixel-based global key features is compensated for by augmented perception of coarse global information, which demonstrates the rationality and effectiveness of the coordinate-attention design. Compared with other pixel-based few-shot segmentation algorithms, the multiscale self-attention mechanism and the coordinate-attention path cooperate in scale fusion and upsampling, respectively. It enables the model to enhance the discriminability between features of the support set and the query set.

TABLE IX

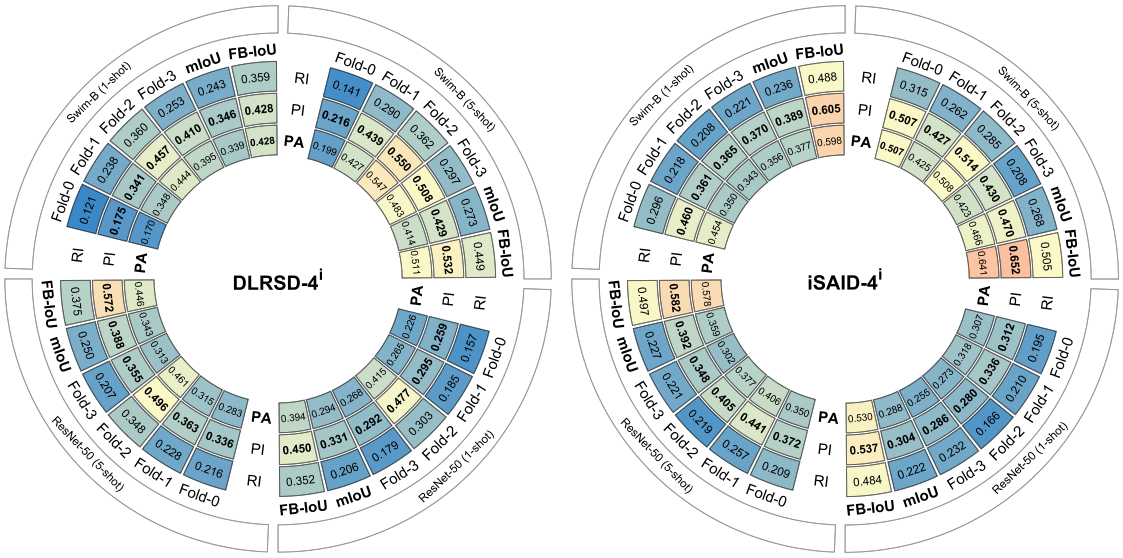
GENERALIZABILITY ANALYSIS (IN %) OF MS^{2}A^{2}NET ON TWO BENCHMARK DATASETS

Dataset	Backbone	1-shot		5-shot	
		mIoU	**FB-IoU**	**mIoU**	**FB-IoU**
DLRSD-4^i	ResNet-50	29.35	39.41	34.27	44.57
iSAID-4^i→DLRSD-4^i		25.39	39.82	27.98	40.36
iSAID-4^i		28.82	52.97	35.86	57.80
DLRSD-4^i→iSAID-4^i		28.37	51.39	36.82	59.00
DLRSD-4^i	ResNet-101	34.53	48.48	39.25	50.90
iSAID-4^i→DLRSD-4^i		32.35	44.09	36.75	45.57
iSAID-4^i		31.18	55.62	35.94	58.48
DLRSD-4^i→iSAID-4^i		31.58	56.41	41.47	62.89
DLRSD-4^i	Swin-B	33.94	42.78	41.38	51.12
iSAID-4^i→DLRSD-4^i		37.13	47.99	44.31	52.40
iSAID-4^i		37.56	59.76	46.57	64.06
DLRSD-4^i→iSAID-4^i		33.29	55.99	45.04	63.60

E. Generalization Analysis

To further demonstrate the generalization of our proposed MS^{2}A^{2}Net, we show its cross-domain performance in Table IX. "DLRSD-4^i→iSAID-4^i" indicates that MS^{2}A^{2}Net is trained on the DLRSD-4^i dataset and tested on the iSAID-4^i dataset, and vice versa. The experimental results show that the generalization performance of moving from small-scale datasets to large-scale datasets is more satisfactory, especially for the ResNet architecture. On the contrary, for the Transformer architecture, the transfer from large-scale to small-scale data is outstanding, which precisely demonstrates that the excellent performance of the Swin-B architecture relies on the large-scale data. Overall, MS^{2}A^{2}Net performs well not only

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(a) (b)

Fig. 7. Ablation studies on the AMCL. (a) DLRSD-4^i dataset. (b) iSAID-4^i dataset. In the figure, "RI," "PI," and "PA" represent the network parameters initialized with random, pre-trained with ImageNet-1K and pre-trained with AMCL, respectively. The best result is represented in bold.

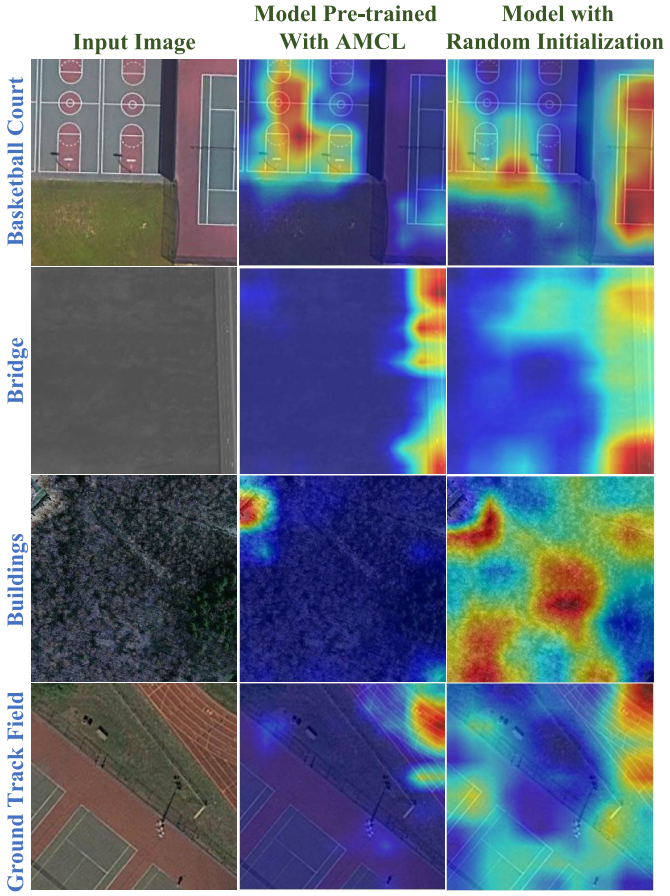


Fig. 8. Heat map visualization comparison of some cases from iSAID-4^i dataset. Each column from left to right represents the input image, the model pre-trained with AMCL and the model random initialization. The heat map visualization results are based on the ResNet-50 backbone.

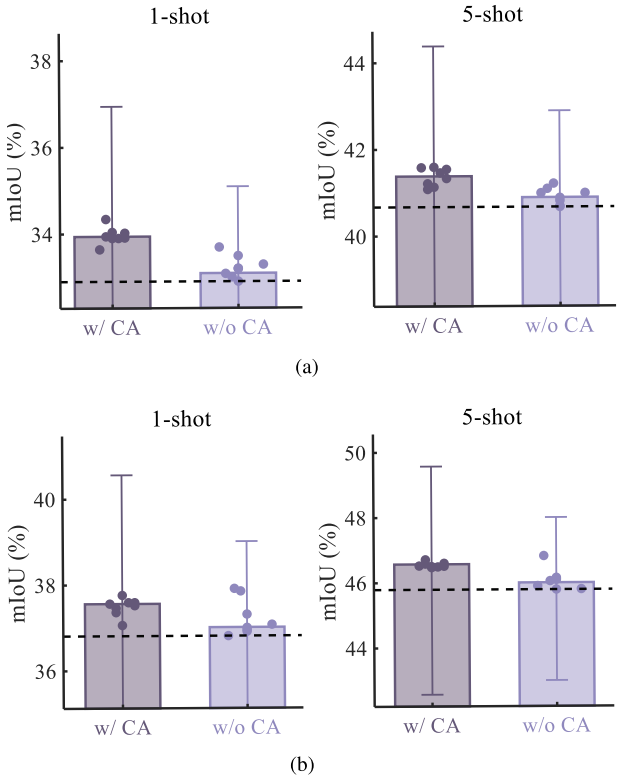


Fig. 9. Ablation experiments on coordinate-attention path. (a) DLRSD-4^i dataset. (b) iSAID-4^i dataset. In the figure, "w/CA" and "w/o CA" denote the inclusion and non-inclusion of coordinate-attention path on the architecture, respectively.

on few-shot remote sensing segmentation, but also in few-shot cross-domain segmentation.

V. Conclusion

In this article, we propose an MS^{2}A^{2}Net for few-shot aerial image segmentation. First, the encoder is pre-trained by AMCL to enhance the discriminability between different segmentation targets. Subsequently, the frozen encoder is transferred to the downstream few-shot segmentation task

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as the feature embedding. In MS^{2}A^{2}Net, the self-attention mechanism is designed to construct a dual connection between foreground and background in the multiscale for query and support features. The coordinate attention mechanism is incorporated in the multiscale feature fusion to enhance the exploration of key information in the global features. In addition, two datasets DLRSD-4^i and iSAID-4^i dedicated to few-shot aerial segmentation are released. The significant performance improvements on these two benchmarks verify the efficacy of the proposed MS^{2}A^{2}Net. It reveals that our approach contributes to the pixel-level foreground and background association in actual remote sensing scenarios and promotes the development of few-shot aerial image segmentation.

REFERENCES

[1] Q. He, X. Sun, W. Diao, Z. Yan, D. Yin, and K. Fu, "RETRACTED: Transformer-induced graph reasoning for multimodal semantic segmentation in remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 90–103, Nov. 2022.

[2] S. Tian, Y. Zhong, Z. Zheng, A. Ma, X. Tan, and L. Zhang, "Large-scale deep learning based binary and semantic change detection in ultra high resolution remote sensing imagery: From benchmark datasets to urban application," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 164–186, Nov. 2022.

[3] L. Ding, H. Tang, Y. Liu, Y. Shi, X. X. Zhu, and L. Bruzzone, "Adversarial shape learning for building extraction in VHR remote sensing images," *IEEE Trans. Image Process.*, vol. 31, pp. 678–690, 2022.

[4] P. Lu, Y. Qin, Z. Li, A. C. Mondini, and N. Casagli, "Landslide mapping from multi-sensor data through improved change detection-based Markov random field," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111235.

[5] X. Lu et al., "Multi-scale and multi-task deep learning framework for automatic road extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9362–9377, Nov. 2019.

[6] Y. Liu, N. Liu, Q. Cao, X. Yao, J. Han, and L. Shao, "Learning non-target knowledge for few-shot semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11563–11572.

[7] C. Lang, G. Cheng, B. Tu, and J. Han, "Few-shot segmentation via divide-and-conquer proxies," *Int. J. Comput. Vis.*, pp. 1–23, 2023, doi: 10.1007/s11263-023-01886-8.

[8] C. Zhang, G. Lin, F. Liu, R. Yao, and C. Shen, "CANet: Class-agnostic segmentation networks with iterative refinement and attentive few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5212–5221.

[9] G. Cheng, C. Lang, and J. Han, "Holistic prototype activation for few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4650–4666, Apr. 2023.

[10] B. Zhang, J. Xiao, and T. Qin, "Self-guided and cross-guided learning for few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8308–8317.

[11] C. Lang, B. Tu, G. Cheng, and J. Han, "Beyond the prototype: Divide-and-conquer proxies for few-shot segmentation," 2022, *arXiv:2204.09903*.

[12] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9196–9205.

[13] C. Lang, G. Cheng, B. Tu, and J. Han, "Learning what not to segment: A new perspective on few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8047–8057.

[14] Z. Tian, H. Zhao, M. Shu, Z. Yang, R. Li, and J. Jia, "Prior guided feature enrichment network for few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 1050–1065, Feb. 2022.

[15] J. Min, D. Kang, and M. Cho, "Hypercorrelation squeeze for few-shot segmenation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6941–6952.

[16] S. Hong, S. Cho, J. Nam, S. Lin, and S. Kim, "Cost aggregation with 4D convolutional Swin transformer for few-shot segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2022, pp. 108–126.

[17] B. Peng et al., "Hierarchical dense correlation distillation for few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23641–23651.

[18] C. Lang, J. Wang, G. Cheng, B. Tu, and J. Han, "Progressive parsing and commonality distillation for few-shot remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613610.

[19] C. Lang, G. Cheng, B. Tu, and J. Han, "Global rectification and decoupled registration for few-shot segmentation in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Aug. 2023, Art. no. 5617211.

[20] A. Shaban, S. Bansal, Z. Liu, I. Essa, and B. Boots, "One-shot learning for semantic segmentation," 2017, *arXiv:1709.03410*.

[21] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2014, pp. 740–755.

[22] Z. Shao, K. Yang, and W. Zhou, "Performance evaluation of single-label and multi-label remote sensing image retrieval using a dense labeling dataset," *Remote Sens.*, vol. 10, no. 6, p. 964, Jun. 2018.

[23] B. Chaudhuri, B. Demir, S. Chaudhuri, and L. Bruzzone, "Multilabel remote sensing image retrieval using a semisupervised graph-theoretic method," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 1144–1158, Feb. 2018.

[24] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 28–37.

[25] B. Wang, Z. Wang, X. Sun, H. Wang, and K. Fu, "DMML-Net: Deep metametric learning for few-shot geographic object segmentation in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3116672.

[26] X. Yao, Q. Cao, X. Feng, G. Cheng, and J. Han, "Scale-aware detailed matching for few-shot aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3119852.

[27] X. Shi et al., "Dense cross-query-and-support attention weighted mask aggregation for few-shot segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 151–168.

[28] L. Madhuanand, F. Nex, and M. Y. Yang, "Self-supervised monocular depth estimation from oblique UAV videos," *ISPRS J. Photogramm. Remote Sens.*, vol. 176, pp. 1–14, Jun. 2021.

[29] S. Saha, P. Ebel, and X. X. Zhu, "Self-supervised multisensor change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022, Art. no. 4405710.

[30] D. Hong et al., "Endmember-guided unmixing network (EGU-Net): A general deep learning framework for self-supervised hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6518–6531, Nov. 2022.

[31] Y. Wang et al., "Self-supervised feature learning with CRF embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2628–2642, May 2019.

[32] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3057768.

[33] C. Tao, J. Qi, W. Lu, H. Wang, and H. Li, "Remote sensing image scene classification with self-supervised paradigm under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 3038420.

[34] Z. Zhao, Z. Luo, J. Li, C. Chen, and Y. Piao, "When self-supervised learning meets scene classification: Remote sensing scene classification based on a multitask learning framework," *Remote Sens.*, vol. 12, no. 20, p. 3276, Oct. 2020.

[35] J. Li, M. Gong, H. Liu, Y. Zhang, M. Zhang, and Y. Wu, "Multiform ensemble self-supervised learning for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3234252.

[36] Z. Zhu, P. Wang, W. Diao, J. Yang, H. Wang, and X. Sun, "Few-shot incremental learning with continual prototype calibration for remote sensing image fine-grained classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 210–227, Feb. 2023.

[37] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1597–1607.

[38] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.

[39] J.-B. Grill et al., "Bootstrap your own latent-a new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21271–21284.

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[40] C. Lang, G. Cheng, B. Tu, C. Li, and J. Han, "Base and meta: A new perspective on few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10669–10686, Sep. 2023.

[41] Y. Zhang, M. Gong, J. Li, M. Zhang, F. Jiang, and H. Zhao, "Self-supervised monocular depth estimation with multiscale perception," *IEEE Trans. Image Process.*, vol. 31, pp. 3251–3266, 2022.

[42] M. Hu, C. Wu, and L. Zhang, "HyperNet: Self-supervised hyperspectral spatial–spectral feature understanding network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3218795.

[43] L. Jian, Z. Pu, L. Zhu, T. Yao, and X. Liang, "SS R-CNN: Self-supervised learning improving mask R-CNN for ship detection in remote sensing images," *Remote Sens.*, vol. 14, no. 17, p. 4383, Sep. 2022.

[44] X. Feng, X. Yao, G. Cheng, J. Han, and J. Han, "SAENet: Self-supervised adversarial and equivariant network for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, Art. no. 5610411.

[45] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[46] H. Li et al., "Global and local contrastive self-supervised learning for semantic segmentation of HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3147513.

[47] W. Li, H. Chen, and Z. Shi, "Semantic segmentation of remote sensing images with self-supervised multitask representation learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6438–6450, 2021.

[48] S. He, Q. Li, Y. Liu, and W. Wang, "Semantic segmentation of remote sensing images with self-supervised semantic-aware inpainting," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] X. Li, D. Shi, X. Diao, and H. Xu, "SCL-MLNet: Boosting few-shot remote sensing scene classification via self-supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3109268.

[50] H. Ji, Z. Gao, Y. Zhang, Y. Wan, C. Li, and T. Mei, "Few-shot scene classification of optical remote sensing images leveraging calibrated pretext tasks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3184080.

[51] T. Gong, X. Zheng, and X. Lu, "Meta self-supervised learning for distribution shifted few-shot scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[52] Q. Zeng and J. Geng, "Task-specific contrastive learning for few-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 143–154, Sep. 2022.

[53] H. Zhou, X. Du, and S. Li, "Self-supervision and self-distillation with multilayer feature contrast for supervision collapse in few-shot remote sensing scene classification," *Remote Sens.*, vol. 14, no. 13, p. 3111, Jun. 2022.

[54] L. Xing, Y. Ma, W. Cao, S. Shao, W. Liu, and B. Liu, "Rethinking few-shot remote sensing scene classification: A good embedding is all you need?" *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[55] N. Inkawhich, "A global model approach to robust few-shot SAR automatic target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[56] Z. Xue, B. Liu, A. Yu, X. Yu, P. Zhang, and X. Tan, "Self-supervised feature representation and few-shot land cover classification of multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3217893.

[57] W. Qiu, Z. Pan, and J. Yang, "Few-shot PolSAR ship detection based on polarimetric features selection and improved contrastive self-supervised learning," *Remote Sens.*, vol. 15, no. 7, p. 1874, Mar. 2023.

[58] W. Zhang, Z. Pan, and Y. Hu, "Exploring PolSAR images representation via self-supervised learning and its application on few-shot classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[59] H. Zhou, X. Du, L. Xia, and S. Li, "Self-learning for few-shot remote sensing image captioning," *Remote Sens.*, vol. 14, no. 18, p. 4606, Sep. 2022.

[60] B. A. A. Monteiro, H. Oliveira, and J. A. d. Santos, "Self-supervised learning for seismic image segmentation from few-labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[61] D. Kang, P. Koniusz, M. Cho, and N. Murray, "Distilling self-supervised vision transformers for weakly-supervised few-shot classification & segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19627–19638.

[62] C. Ouyang, C. Biffi, C. Chen, T. Kart, H. Qiu, and D. Rueckert, "Self-supervised learning for few-shot medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 41, no. 7, pp. 1837–1848, Jul. 2022.

[63] C. Sharma and M. Kaul, "Self-supervised few-shot learning on point clouds," *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 7212–7221.

[64] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[65] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15745–15753.

[66] Z. Xie et al., "Self-supervised learning with Swin transformers," 2021, *arXiv:2105.04553*.

[67] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[68] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.

[69] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13708–13717.

[70] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[71] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[72] V. Mnih, "Machine learning for aerial image labeling," Ph.D. dissertation, Dept. Comput. Sci., Univ. Toronto, Toronto, ON, Canada, 2013.

[73] X. Jiang, N. Zhou, and X. Li, "Few-shot segmentation of remote sensing images using deep metric learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[74] G. Puthumanaillam and U. Verma, "Texture based prototypical network for few-shot semantic segmentation of forest cover: Generalizing for different geographical regions," *Neurocomputing*, vol. 538, Jun. 2023, Art. no. 126201.



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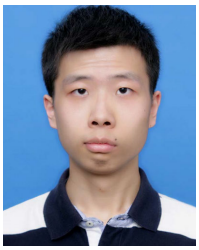
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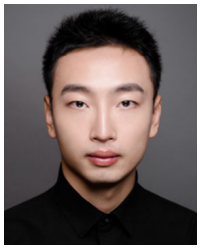
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